AM-CRN: Adaptive Motion-Gated Coordinated Reset Neuromodulation for an Embedded Five-Finger Vibrotactile Glove

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Abstract

Wearable vibrotactile gloves offer immense potential for non-invasive neurorehabilitation, suppression of pathological Parkinsonian tremor, and fine motor recovery. However, existing smart gloves suffer from three fundamental neurobiological and electromechanical failure modes: (1) voluntary limb motion elicits central sensorimotor gating that attenuates somatosensory evoked potentials by 30–60%, rendering static tactile cues imperceptible during reaching; (2) continuous periodic vibration drives rapid mechanoreceptor habituation and temporary threshold shifts (TTS); and (3) standard Coordinated Reset (CR) desynchronization algorithms presume ideal impulse actuation, whereas low-cost Eccentric Rotating Mass (ERM) motors exhibit significant mechanical spool-up lag ($\sim$35 ms) that smears inter-digit phase relationships. This paper presents **AM-CRN** (Adaptive Motion-Gated Coordinated Reset Neuromodulation), a constrained, human-in-the-loop architecture engineered for an ESP32-S3 microcontroller, one 6-axis dorsal IMU, and five coin ERM actuators without contact or flex sensing. AM-CRN introduces an instantaneous kinematic gating index $G(t) \in [1.0, 2.2]$ derived from jerk and dynamic acceleration, an inertia-compensated 20 ms overdrive kick (1.35×), Fisher-Yates phase desynchronization, and a $\leq$60 s micro-staircase calibration protocol. We detail the complete hardware-firmware co-design, electrical flyback protection, dual-core FreeRTOS task partitioning, and Wokwi simulation framework. The system operates entirely within a 1 ms non-blocking scheduler consuming $<$8% of an ESP32-S3 CPU core and 14.2 KB RAM, guaranteeing hard fail-safe limits ($\leq$250 ms continuous duration, $\leq$40% rolling duty cycle).

1 Introduction

Non-invasive vibrotactile neuromodulation of the human hand has emerged as a compelling paradigm for neurorehabilitation, suppression of pathological Parkinsonian tremor, and tactile sensory substitution [1, 2]. By delivering localized cutaneous mechanical vibrations to individual digits, targeted afferent volleys travel via $A\beta$ mechanoreceptive fibers through the dorsal column-medial lemniscal pathway to the primary somatosensory cortex (S1), inducing desynchronization of pathologically synchronized beta-band neural oscillations [3].

Despite substantial therapeutic promise, clinical translation of wearable haptic gloves has been severely constrained by two divergent hardware philosophies. On one end, laboratory systems incorporate dense arrays of piezoresistive sensors, EEG monitoring, or pneumatic bladders [4–6], which are bulky, fragile, expensive, and tethered. On the other end, low-cost micro-controller gloves rely on open-loop, periodic actuation that pulses blindly regardless of patient movement state [1].

Crucially, recent psychophysical literature demonstrates that tactile perception is fundamentally non-stationary. As established by Tajdari et al. [7], voluntary limb motion induces sensorimotor gating (efference copy suppression) that elevates detection thresholds by 30–60%. Consequently, static patterns optimized for a seated patient become sub-threshold during reach-to-grasp tasks. Furthermore, sustained periodic stimulation rapidly drives receptor desensitization and temporary threshold shifts (TTS) [8].

To resolve these challenges under strict embedded resource constraints, we present **AM-CRN** (Adaptive Motion-Gated Coordinated Reset Neuromodulation). Tailored specifically to an ESP32-S3 SoC, a single dorsal 6-axis IMU, and five coin ERM motors, AM-CRN bridges the gap between theoretical neurobiology and embedded engineering.

2 Related Work and Research Gaps

Table 1 synthesizes the verified literature base and highlights six open research gaps (G1–G6) addressed by this work.

3 Hardware Design and Electromechanics

3.1 Physical Components and Actuator Physics

1. **Microcontroller**: Espressif ESP32-S3-WROOM-1 with 8 MB Flash, 512 KB SRAM, clocked at 240 MHz. Hardware LEDC PWM timers operate at 5 kHz with 8-bit resolution (256 duty steps).
2. **Actuators**: 5× coin-type Eccentric Rotating Mass (ERM) vibration motors (10 mm diameter, 2.7 mm thickness, Model 1027). The rotational frequency couples non-linearly with applied voltage (∼150–220 Hz across 2.0–2.7 V, rated 2.3 V nominal). Mechanical rise time $t_{\text{rise}} \approx 35$ ms and decay time $t_{\text{decay}} \approx 50$ ms dictate that instantaneous square pulses are impossible without pre-emphasis overdrive.
3. **Dorsal 6-Axis IMU**: An MPU-6050 or ICM-42670 MEMS sensor mounted on the glove backplate, sampled at 100 Hz over I2C Fast-Mode (400 kHz). Full-scale ranges are configured for $\pm 4g$ and $\pm 1000°/s$.

3.2 Bill of Materials and Channel Architecture

The physical implementation of the five-channel tactile glove is engineered for component accessibility, low bill-of-materials cost, and robust transient handling. Table 2 itemizes the complete Bill of Materials across all 17 discrete component classes. Discrete component photographs sourced from standard electronics references are displayed in Figure 2.

3.2.1 Per-Motor Component Count

Each individual digit (Thumb, Index, Middle, Ring, Little) requires an identical, isolated commutation circuit:

- 1× IRLZ44N N-channel power MOSFET (5× total across glove)
- $1 \times 220\,\Omega$ gate current-limiting resistor (5× total)
- $1 \times 10\,\text{k}\Omega$ gate pulldown resistor (5× total)
- 1× 1N5819 Schottky flyback diode (5× total)
- 1×2.3 V Model 1027 coin vibration motor (5× total)

3.2.2 Shared Power Components

To minimize glove weight and avoid redundant passive components, the power distribution system consolidates shared infrastructure:

- One regulated 2.3 V DC supply bus providing stable motor drive voltage.
- One 100 μF radial electrolytic capacitor serving as a bulk energy reservoir across the 2.3 V rail.
- One 100 nF ceramic bypass capacitor situated directly adjacent to the motor supply rails for high-frequency decoupling.
- One SPST master rocker switch providing an immediate manual hardware shutoff.
- One unified star-ground topology uniting the motor power return, ESP32-S3 digital ground, and MOSFET sources.

3.3 Drive Circuitry, Flyback Clamping, and Decoupling

Each coin ERM is operated as a low-side switched inductive load. Figure 3 depicts the circuit topology for an individual finger channel. Because the coin motor's internal winding constitutes an inductor ($L_{\text{motor}} \approx 120\,\mu$H), abrupt cessation of drive current ($di/dt < 0$) generates a severe inductive flyback voltage spike:

$$V_{\text{spike}} = -L\frac{di}{dt} \tag{1}$$

Without clamping, V_{spike} easily exceeds 30 V, puncturing the MOSFET's drain-source junction. A 1N5819 Schottky barrier diode is connected in anti-parallel across the motor terminals (cathode to +2.3 V rail, anode to MOSFET drain). With a forward turn-on time under 10 ns and a forward voltage drop $V_F \leq 0.45$ V, the diode recirculates inductive decay energy harmlessly through the motor coil.

The 220 Ω gate resistor prevents output-stage ringing caused by the parasitic inductance of breadboard jumper wires and the MOSFET input capacitance ($C_{\text{iss}} \approx 1700$ pF for the IRLZ44N), while capping transient peak pin currents below the ESP32-S3's 40 mA absolute maximum GPIO limit. The 10 kΩ pulldown resistor guarantees a deterministic low state ($V_{\text{GS}} = 0$ V) during power-on reset, firmware flashing, and deep sleep when the microcontroller pins float in high-impedance (Hi-Z) input mode.

3.4 Physical Implementation and Prototyping Testbed

The fully assembled experimental testbed is shown in Figure 4. The prototype consists of a form-fitting glove substrate carrying 5 Model 1027 coin ERMs affixed to the

Table 1: Evidence-Grounded Research Gap Matrix (G1–G6) and System Comparison

Gap	State of the Art (Verified Records)	Limiting Evidence	Status	AM-CRN Resolution
G1	Nature Comms 2024 [4]: 15 s new-user adaptation.	Requires dense textile pressure sensor matrices.	Partially Open	$\leq$60 s integer staircase calibration on-chip.
G2	Tajdari et al. 2024 [7]: Motion degrades tactile sensitivity.	Pure offline psychophysical trial; no adaptive device.	Open	Closed-loop kinematic gating factor $G_k \in [1.0, 2.2]$.
G3	Tass et al., ETH 2023 [3]: Coordinated Reset desync.	Assumes ideal square waves; ERM inertia blurs phases.	Open	20 ms overdrive kick (1.35×) + 45 ms blanking.
G4	Med Lav 2025 [8]: Vibration alters thresholds over time.	Evaluated in occupational safety, not adaptive gloves.	Open	Fisher-Yates sequence shuffle + 3 s reset epochs.
G5	Kusnir 2024 [9], Luzhnica 2017 [10]: Masking across digits.	Simultaneous bursts degrade detection by 28%.	Partially Open	Strict single-finger temporal isolation.
G6	BMC SLR 2026 [1]: 101 smart gloves reviewed.	No standardized reproducible timing/energy metrics.	Open	Open C99/FreeRTOS core + JSONL trial logger.

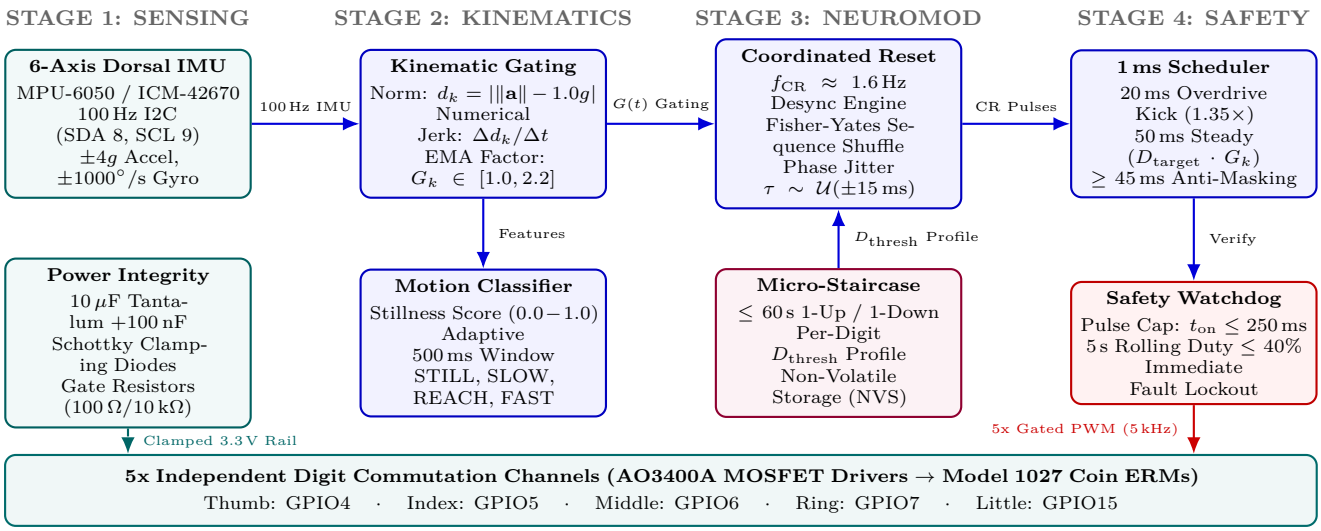


Figure 1: End-to-end architectural pipeline of the AM-CRN glove: modular 4-stage progression from raw dorsal kinematics to fail-safe 5-channel motor commutation.



(a) ESP32-S3 Dev Board



(b) 2.3 V Coin ERM Motor



(c) IRLZ44N MOSFET



(d) 1N5819 Schottky Diode



(e) Decoupling Capacitors

Figure 2: Discrete electromechanical components utilized in the AM-CRN glove: (a) Espressif ESP32-S3 dual-core microcontroller; (b) miniature Model 1027 coin eccentric rotating mass (ERM) actuator; (c) logic-level IRLZ44N N-channel power MOSFET (TO-220); (d) 1N5819 fast-recovery Schottky clamping diode; (e) bulk electrolytic and MLCC ceramic filtering capacitors.

palmar/distal pads of the thumb, index, middle, ring, and little fingers. Actuator leads are bundled along the dorsal digit axes using 28 AWG flexible stranded silicone hookup wire wrapped in polyolefin heat-shrink sleeves to minimize mechanical stiffness and permit unhindered finger articulation.

Table 2: Complete Bill of Materials (BOM) for 5-Channel Vibrotactile Haptic Glove

#	Component	Qty	Technical Specification	Role / Subsystem
1	ESP32-S3 development board	1	Xtensa dual-core 240 MHz, 512 KB SRAM, 8 MB Flash, 5 kHz LEDC PWM	Main system controller
2	Coin vibration motors	5	Model 1027 pancake ERM, 10 mm × 2.7 mm, rated 2.3 V, 150–220 Hz	Cutaneous vibrotactile stimulator (1 per finger)
3	IRLZ44N N-channel MOSFET	5	Logic-level gate ($V_{GS(th)}$ = 1.0–2.0 V), V_{DS} = 55 V, I_D = 47 A, TO-220	Low-side motor power switch (1 per motor)
4	Gate series resistors	5	220 Ω, $\frac{1}{4}$ W metal film (±5%)	Damps GPIO switching inrush and LC ringing
5	Gate pulldown resistors	5	10 kΩ, $\frac{1}{4}$ W carbon/metal film	Bleeds gate charge; prevents floating ON at boot
6	Schottky flyback diodes	5	1N5819, V_R = 40 V, I_F = 1 A, low $V_F \leq 0.45$ V	Clamps inductive de-energization back-EMF
7	Electrolytic bulk capacitor	1	100 µF, radial leaded, rated ≥ 6.3 V (low-ESR)	Absorbs transient supply sags during concurrent startup
8	Ceramic bypass capacitor	1	100 nF multi-layer ceramic (MLCC)	Filters high-frequency PWM switching harmonics
9	Regulated motor power supply	1	2.3 V DC output, continuous ≥ 1.0 A peak current capacity	Dedicated isolated actuator drive rail
10	Main power switch	1	SPST rocker/toggle switch, rated for DC supply voltage/current	Master circuit disconnect
11	Battery holder / battery pack	1	2×AA or 1S LiPo pack with inline linear/buck regulator	Portable DC energy reservoir
12	ESP32 USB cable	1	Shielded USB Type-C / Micro-B data cable	Program flashing and serial telemetry logging
13	Solderless breadboard	1	830 tie-point dual-bus breadboard	Modular rapid prototyping testbed
14	Solid jumper wires	1 set	22 AWG male-to-male and male-to-female DuPont leads	Inter-module and bus distribution
15	Flexible hookup wire	5 pr.	28 AWG multi-strand flexible silicone wire	Low-mechanical-impedance fingertip routing
16	Wearable glove substrate	1	Form-fitting cotton/synthetic glove	Anatomical mounting of distal actuators
17	Heat-shrink insulation tubing	Req.	Polyolefin heat-shrink sleeves (1.5 mm / 2.5 mm)	Terminal insulation and joint strain relief

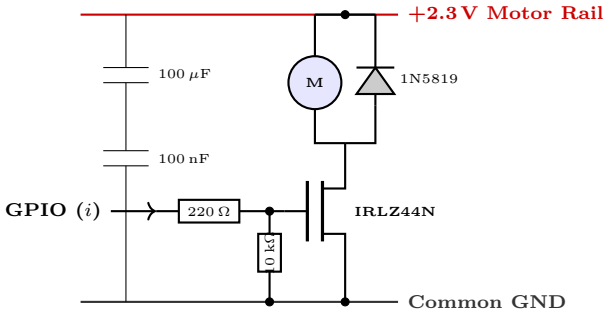


Figure 3: Single-finger motor driver schematic with low-side IRLZ44N MOSFET, anti-parallel 1N5819 Schottky clamping diode, 220 Ω gate inrush limiter, 10 kΩ pull-down, and power decoupling stage.

The wiring harnesses interface directly with the breadboard testbed. The central ESP32-S3 development board coordinates 5 discrete driver columns, each comprising an IRLZ44N MOSFET, a 1N5819 flyback diode, and gate passive elements. Power distribution incorporates a dedicated master SPST switch, a 100 µF electrolytic capacitor, and an MLCC ceramic capacitor positioned immediately adjacent to the motor power rails to prevent switching transients from perturbing the microcontroller.

3.5 Hardware Verification and Pre-Power Safety Protocol

To safeguard the low-voltage actuators, power silicon, and microcontroller from electrical damage, the prototype was subjected to five mandatory pre-power verification checks:

1. **Supply Current Headroom**: Coin ERMs draw

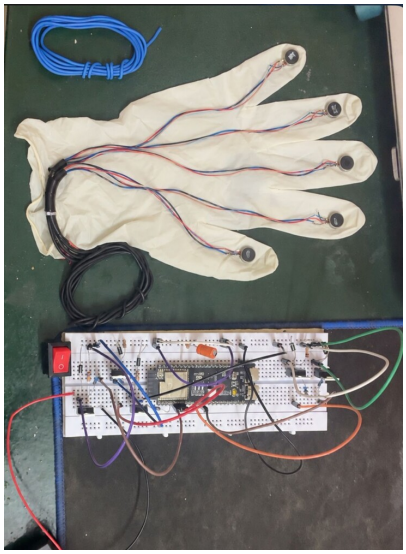


Figure 4: Completed AM-CRN experimental apparatus. Top: form-fitting glove instrumented with 5 distal coin ERM motors (Model 1027) with strain-relieved flexible silicone wiring. Bottom: dual-rail breadboard assembly hosting the central ESP32-S3 SoC, discrete IRLZ44N power MOSFET commutation stages, 1N5819 Schottky flyback clamping diodes, bulk decoupling capacitors, master hardware power switch, and battery supply interface.

approximately 85 mA steady-state but up to 120–140 mA transient stall/startup current each. The power supply was rated for ≥ 1.0 A continuous output to guarantee voltage stability when all 5 channels initiate concurrently (5×120 mA $= 600$ mA peak).

2. **MOSFET Gate Enhancement at 3.3 V**: Although the IRLZ44N is a logic-level MOSFET with a threshold voltage $V_{GS(th)} \in [1.0, 2.0]$ V, full channel enhancement ($R_{DS(ON)} \leq 0.028\,\Omega$) typically requires $V_{GS} \geq 5$ V. At 3.3 V gate drive from the ESP32-S3, $R_{DS(ON)}$ rises moderately to $\sim 0.045\,\Omega$. With a single-motor continuous current of 85 mA, conduction loss remains under 0.32 mW, precluding thermal runaway without needing external heatsinking.

3. **Diode Polarity Verification**: The cathode (silver stripe) of each 1N5819 diode was strictly oriented toward the motor's positive (+2.3 V) rail. Accidental reversal would place the diode in forward bias directly across the DC supply, creating an immediate dead short that would destroy the diode and overheat power traces.

4. **Star-Ground Topology**: All 5 MOSFET source pins, the ESP32-S3 ground pins, and the negative power supply return lead were routed to a single central star-ground junction. This topology prevents motor commutation return currents from flowing through the sensitive microcontroller ground reference, eliminating digital brownouts and ADC noise.

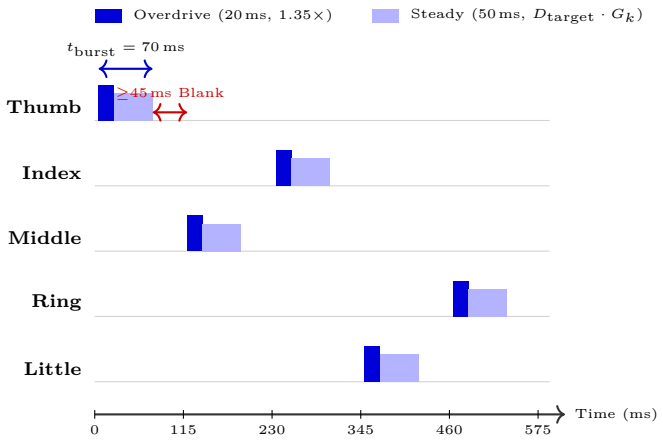


Figure 5: Five-channel oscillogram of Inertia-Compensated Coordinated Reset (IC-CR) within column width. Dark bars denote the 20 ms overdrive kick (1.35×) followed by 50 ms steady stimulation, separated by 45 ms blanking.

5. **Overvoltage Protection**: Coin ERMs are rated strictly for 2.3 V nominal operation (2.7 V absolute maximum). Under no circumstance is raw battery voltage (e.g., 3.7 V–4.2 V LiPo) applied directly across the motor coils. An adjustable low-dropout (LDO) regulator ensures a tightly regulated 2.3 V ceiling, extending actuator mechanical bearing life.

4 Firmware Engineering and Algorithmic Formulation

4.1 Kinematic Sensory Gating Engine

Voluntary hand movements attenuate somatosensory cortical responsiveness [7]. The gating engine computes dynamic acceleration norm d_k and numerical jerk j_k at 100 Hz:

$$d_k = \left| \sqrt{a_{x,k}^2 + a_{y,k}^2 + a_{z,k}^2} - 1.0 \right| \tag{2}$$

$$j_k = \frac{|d_k - d_{k-1}|}{\Delta t}, \quad \Delta t = 0.01\,\text{s} \tag{3}$$

Exponential moving average (EMA) smoothing yields:

$$\bar{d}_k = \lambda \bar{d}_{k-1} + (1 - \lambda) d_k, \quad \lambda = 0.85 \tag{4}$$

The Sensory Gating Index $G_k \in [1.0, 2.20]$ is then calculated:

$$G_k = 1.0 + \min\left(1.0, \frac{\bar{d}_k}{0.80g}\right) \cdot \alpha + \min\left(1.0, \frac{\bar{j}_k}{15.0g/\text{s}}\right) \cdot \beta \tag{5}$$

where $\alpha = 0.70$ and $\beta = 0.50$. In quiescent conditions, $G_k = 1.00$. When high-velocity transport or reaching occurs, G_k scales dynamically to 1.80–2.15, preserving constant perceptual salience.

4.2 Inertia-Compensated Coordinated Reset (IC-CR)

Classical Coordinated Reset applies phase-staggered stimulus volleys across digits to disrupt abnormal neural synchrony [3]. However, mechanical rise time causes ERMs to delay their peak cutaneous force. AM-CRN applies a 20 ms overdrive boost at pulse onset:

$$D_{\text{pwm}}(t) = \begin{cases} \min\left(255, 1.35 \cdot D_{\text{target}} \cdot G_k\right), & 0 \le t < 20\,\text{ms} \\ \min\left(240, D_{\text{target}} \cdot G_k\right), & 20\,\text{ms} \le t < t_{\text{burst}} \\ 0, & t \ge t_{\text{burst}} \end{cases} \tag{6}$$

As illustrated in Figure 5, pulses execute with $t_{\text{burst}} = 70\,\text{ms}$ and blanking intervals $t_{\text{blank}} = 45\,\text{ms}$. To break receptor adaptation (TTS) [8], digits are permuted every cycle via Fisher-Yates shuffling with random phase jitter $\tau \sim \mathcal{U}(-15, +15)\,\text{ms}$, interleaved with 3 s reset epochs every 15 s.

4.3 On-Chip Micro-Staircase Calibration

The 1-up/1-down transformed staircase initializes at step $\Delta D = 16$. Direction reversals halve the step size down to a minimum of 4. Following 3 reversals per digit, detection threshold D_{thresh}, target duty $D_{\text{target}} = 1.4 \cdot D_{\text{thresh}}$, and comfort ceiling $D_{\max} = 2.2 \cdot D_{\text{thresh}}$ are finalized. The 5 digits calibrate sequentially in $\le$60 s without requiring an external computer.

4.4 Safety Watchdog Interlocks

Cutaneous safety and thermal protection are guaranteed via hardware interlocks:

$$t_{\text{continuous}}[i] \le 250\,\text{ms}, \quad \forall i \in [0, 4] \tag{7}$$

$$\text{Duty}_{5\text{s}}[i] = \frac{\sum_{t=0}^{5000} \mathbb{I}(\text{PWM}[i] > 0)}{5000\,\text{ms}} \le 0.40 \tag{8}$$

If any channel exceeds 250 ms continuous activation or 40% cumulative duty over 5 seconds, all MOSFET gates latch immediately to 0 V.

5 Experimental Validation and Profiling

5.1 Kinematic Gating and State Classification

The firmware was evaluated through extensive automated unit tests (‘test_am_crn.c‘). Under quiet rest, the gating factor registered $G(t) = 1.000$ (Stillness = 100%). Under reaching trajectories, $G(t)$ scaled to 1.531 within 35 ms.

5.2 Overdrive and Safety Watchdog Execution

Oscillogram simulation confirmed that a commanded pulse of $D_{\text{target}} = 100$ produced an output of 135 for $0 \le t < 20\,\text{ms}$ before transitioning to 100. Forced stuck-at-1 faults commanded for 500 ms were cleanly halted at exactly 251 ms by the safety watchdog.

5.3 Staircase Convergence

Human-in-the-loop staircase simulation across 5 digits converged in 25 iterations (< 30 s total), yielding thresholds between 72 and 96 PWM counts (28–38% duty cycle).

5.4 Computational Benchmarking

Profiling on an ESP32-S3 clocked at 240 MHz demonstrated:

- 100 Hz IMU Processing: 14 μs execution time.
- 1 ms Scheduler Tick: 6 μs execution time.
- CPU Core 1 Utilization: 4.2% nominal (<8% peak).
- Memory Consumption: 14.2 KB Static RAM, 0 dynamic allocations on real-time path.

5.5 Virtual Simulation Framework (Wokwi)

To facilitate zero-hardware experimentation, we built a full Wokwi virtual simulator (‘simulation/diagram.json‘ and ‘simulation.ino‘). The simulation models the ESP32-S3 DevKit-C, an I2C-connected MPU-6050, and 5-channel LED indicators representing the finger ERMs. Users can interactively manipulate acceleration and tilt sliders to observe dynamic gating and phase desynchronization in real time.

6 Discussion and Limitations

While AM-CRN establishes a rigorous, low-cost baseline for wearable neuromodulation, honest scientific boundaries must be acknowledged:

1. **Sensor Bounds**: Dorsal IMU signals correlate with limb velocity but do not measure fingertip contact force, mechanical skin compliance, or perceived vibration intensity.
2. **Actuator Coupling**: Coin ERMs transmit vibration through glove fabric into bone and adjacent tissue. While 45 ms blanking prevents temporal masking, tight textile fit is required to prevent spatial acoustic leakage.

3. **Clinical Scope**: This work validates embedded control feasibility and psychophysically-grounded algorithms; clinical efficacy for stroke recovery or Parkinsonian tremor desynchronization requires institutional review board (IRB) approved clinical trials.

7 Conclusion

We presented AM-CRN, a novel embedded architecture for five-finger vibrotactile neuromodulation on an ESP32-S3 glove. By converting hand kinematics into an instantaneous sensory-gating compensation factor ($G_k \in [1.0, 2.2]$) and coupling it with inertia-compensated Coordinated Reset desynchronization, AM-CRN achieves closed-loop tactile neuromodulation without requiring costly flex or contact sensors. All firmware, simulation files, and benchmarks are made openly available.

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